

Advanced Quantum Mechanics - Review Mock Exam

Based on Lecture 15 Review

Winter Semester 2025/2026

Instructions

- This exam focuses on the theoretical concepts from Lectures 1 to 14.
 - Each question has **5 options**.
 - **Any number of options (from 0 to 5)** may be correct for a given question.
 - Select all statements that are physically and mathematically correct.
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1. The Schrödinger Equation and Time Evolution

Consider the general formalism of quantum mechanics as reviewed in the slides. Which of the following statements are correct?

- A. The time-evolution operator $\hat{U}(t) = e^{-\frac{i}{\hbar}\hat{H}t}$ is unitary if and only if the Hamiltonian $\hat{H}$ is Hermitian.
- B. For a time-independent Hamiltonian, the probability density $|\psi(x, t)|^2$ of a stationary state $|E_n\rangle$ changes sinusoidally with time.
- C. The general solution to the time-dependent Schrödinger equation can be expanded as a superposition of energy eigenstates: $|\psi(t)\rangle = \sum_n c_n(0)e^{-iE_n t/\hbar}|E_n\rangle$.
- D. The norm of the state vector $\langle\psi(t)|\psi(t)\rangle$ is conserved during unitary time evolution.
- E. In the position representation, the momentum operator acts as a multiplication operator ($\hat{p} \rightarrow p$).

Correct Answers: A, C, D

2. The Quantum Harmonic Oscillator

Regarding the 1D quantum harmonic oscillator and its algebraic solution using ladder operators $\hat{a}$ and $\hat{a}^\dagger$:

- A. The energy spectrum is equidistant with $E_n = \hbar\omega(n + 1/2)$.
- B. The ground state energy is zero ($E_0 = 0$).
- C. The creation operator $\hat{a}^\dagger$ increases the particle number/excitation level by 1, such that $\hat{a}^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle$.
- D. The position operator $\hat{x}$ can be written as a linear combination of $\hat{a}$ and $\hat{a}^\dagger$.
- E. The ground state wavefunction $\langle x|0\rangle$ is an odd function of x (antisymmetric).

Correct Answers: A, C, D

3. The Hydrogen Atom

Consider the solution to the Hydrogen atom problem ($V(r) \propto -1/r$):

- A. The energy eigenvalues E_n depend only on the principal quantum number n (ignoring fine structure).
- B. The degeneracy of the energy level E_n is n^2 (including spin degeneracy, it would be $2n^2$, but neglecting spin it is n^2).
- C. The wavefunctions are factorized into a radial part $R_{nl}(r)$ and spherical harmonics $Y_{lm}(\theta, \phi)$.
- D. The $l = 0$ (s-orbitals) have non-zero probability density at the nucleus ($r = 0$).
- E. The centrifugal barrier term in the radial equation vanishes for $l = 0$.

Correct Answers: A, B, C, D, E

4. Approximation Methods (Perturbation Theory)

Which of the following statements regarding time-independent and time-dependent perturbation theory are true?

- A. The first-order energy correction in time-independent perturbation theory is the expectation value of the perturbation in the unperturbed state: $E_n^{(1)} = \langle n^{(0)} | \hat{V} | n^{(0)} \rangle$.
- B. Fermi's Golden Rule gives the transition rate for a system interacting with a continuum of final states under a constant or harmonic perturbation.
- C. Second-order perturbation theory always leads to an increase in the ground state energy.
- D. To apply non-degenerate perturbation theory, the energy difference between states must be large compared to the matrix element of the perturbation.
- E. In the interaction picture, the state vector is constant in time.

Correct Answers: A, B, D

5. Scattering Theory and Partial Waves

Regarding scattering by a spherically symmetric potential:

- A. The First Born approximation is generally most accurate at very low incident energies ($k \rightarrow 0$).
- B. In the low-energy limit ($kR \ll 1$), s-wave ($l = 0$) scattering usually dominates the cross-section.
- C. The differential cross-section is related to the scattering amplitude by $d\sigma/d\Omega = |f(\theta)|^2$.
- D. Phase shifts $\delta_l(k)$ are necessary to calculate the scattering amplitude in the partial wave method.
- E. The scattering amplitude $f(\theta)$ is purely real for all potentials.

Correct Answers: B, C, D

6. Many-Body Physics and Identical Particles

Consider a system of N identical particles:

- A. For bosons, the total wavefunction must be antisymmetric under the exchange of any two particles.
- B. Fermions obey the Pauli exclusion principle, meaning no two fermions can occupy the same single-particle state.
- C. The ground state of two non-interacting bosons in a harmonic trap has energy $E_{gr} = \hbar\omega$.
- D. The exchange symmetry requirement arises because identical particles are indistinguishable in quantum mechanics.
- E. The Slater determinant is used to construct symmetric wavefunctions for bosons.

Correct Answers: B, C, D

7. H_2^+ and Helium Atom

Regarding the application of quantum mechanics to simple molecules and multi-electron atoms:

- A. The Born-Oppenheimer approximation relies on the fact that the nuclei are much heavier (and thus move slower) than the electrons.
- B. In the H_2^+ molecule, the bonding orbital is antisymmetric with respect to the inversion of coordinates.
- C. The ground state of the Helium atom is a spin singlet state (para-helium).
- D. The Hartree-Fock method captures all electron-electron correlation effects perfectly.
- E. The variational method states that any trial wavefunction yields an energy expectation value greater than or equal to the true ground state energy.

Correct Answers: A, C, E

8. Second Quantization formalism

In the formalism of second quantization:

- A. It is a necessary framework to describe systems where the number of particles can change.
- B. Bosonic creation and annihilation operators satisfy commutation relations $[\hat{a}_i, \hat{a}_j^\dagger] = \delta_{ij}$.
- C. Fermionic field operators satisfy anticommutation relations $\{\hat{\Psi}(\mathbf{x}), \hat{\Psi}^\dagger(\mathbf{y})\} = \delta(\mathbf{x} - \mathbf{y})$.
- D. The operator $\hat{N} = \sum_i \hat{a}_i^\dagger \hat{a}_i$ counts the total number of particles in the system.
- E. A two-body interaction term involves four operators (e.g., $\hat{a}^\dagger \hat{a}^\dagger \hat{a} \hat{a}$).

Correct Answers: A, B, C, D, E